

Complex Integration

4410

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Cauchy's Integral Formula

Theorem (Cauchy Integral Formula).

Let f be analytic on and inside a simple closed positively oriented contour γ . Let z_0 be a point inside γ . Then

$$f(z_0) = \frac{1}{2\pi i} \int_{\gamma} \frac{f(z)}{z - z_0} dz.$$

Why this is astonishing:

- The value of f at z_0 is completely determined by its values on the boundary.
- Analytic functions are incredibly rigid.
- Integration produces evaluation.

Holomorphic functions are controlled by their boundary behavior.

Cauchy Formula for Derivatives

Theorem (Higher Derivative Formula).

If f is analytic on and inside γ and z_0 is inside γ , then for $n \geq 1$:

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \int_{\gamma} \frac{f(z)}{(z - z_0)^{n+1}} dz.$$

Why this is astonishing:

- Analytic $\Rightarrow$ infinitely differentiable.
- Derivatives are given by integral formulas.
- Leads to power series expansions:

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n.$$

Differentiation becomes a byproduct of integration.

Riemann Integral: Limit Definition

Let $f : [a, b] \rightarrow \mathbb{R}$ be bounded. Take a partition

$$P = \{a = x_0 < x_1 < \cdots < x_n = b\}$$

and choose sample points $\xi_i \in [x_{i-1}, x_i]$. Then the **Riemann sum** is

$$S(f, P) = \sum_{i=1}^n f(\xi_i) \Delta x_i, \quad \Delta x_i = x_i - x_{i-1}.$$

If the limit exists the Riemann integral is

$$\int_a^b f(x) dx := \lim_{\|P\| \rightarrow 0} S(f, P).$$

Alternate version with $m_i = \inf_{[x_{i-1}, x_i]} f$ and $M_i = \sup_{[x_{i-1}, x_i]} f$. If

$$\sup_P L(f, P) = \inf_P U(f, P).$$

then f is Riemann integrable where

$$L(f, P) = \sum m_i \Delta x_i, \quad U(f, P) = \sum M_i \Delta x_i,$$

Properties of the Riemann Integral

- Linearity: $\int_a^b (cf + g) = c \int_a^b f + \int_a^b g$.
- Monotonicity: $f \leq g \Rightarrow \int_a^b f \leq \int_a^b g$.
- Additivity: $\int_a^b f = \int_a^c f + \int_c^b f$.
- Continuous f are Riemann integrable.

Fundamental Theorem of Calculus (Calc II)

Let f be continuous on $[a, b]$ and define

$$F(x) = \int_a^x f(t) dt.$$

- Then $F'(x) = f(x)$ for all $x \in [a, b]$.
- For any antiderivative F of f ,

$$\int_a^b f(x) dx = F(b) - F(a).$$

This theorem connects differentiation and integration and is the basis for evaluating definite integrals.

Green's Theorem

$$\oint_C (P \, dx + Q \, dy) = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

for positively oriented simple closed curve C and D its interior.

Interpretation: circulation around C = double integral of curl over D .

Example (Green's Theorem)

$\oint_C (x^2 dx + y^2 dy)$ around unit circle:

$$\iint_D \left(\frac{\partial}{\partial x} y^2 - \frac{\partial}{\partial y} x^2 \right) dA = 0$$

Divergence Theorem (Gauss)

$$\iint_S \mathbf{F} \cdot \mathbf{n} \, dS = \iiint_V \nabla \cdot \mathbf{F} \, dV$$

for vector field $\mathbf{F}$ on solid V with boundary S .

Stokes' Theorem

$$\oint_{\partial S} \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}.$$

Relates circulation along a boundary to curl over surface.

Complex Integrals (Section 6.1)

Let $f : [a, b] \rightarrow \mathbb{C}$, $f(t) = u(t) + iv(t)$. Define the integral by

$$\int_a^b f(t) dt := \int_a^b u(t) dt + i \int_a^b v(t) dt.$$

Riemann sum definition:

$$\int_a^b f(t) dt = \lim_{\|P\| \rightarrow 0} \sum_{i=1}^n f(\xi_i) \Delta t_i.$$

Chopping up explanation: Partition $[a, b]$ into small intervals. On each interval $[t_{i-1}, t_i]$, approximate the integral by $f(\xi_i) \Delta t_i$. Sum over all intervals. Take the limit as the maximum interval length $\|P\| \rightarrow 0$.

Contours and Contour Integrals (Section 6.2)

A contour $\gamma : [a, b] \rightarrow \mathbb{C}$, $\gamma(t) = x(t) + iy(t)$, defines

$$\int_{\gamma} f(z) dz := \int_a^b f(\gamma(t)) \gamma'(t) dt.$$

Riemann sum version:

$$\int_{\gamma} f(z) dz = \lim_{\|P\| \rightarrow 0} \sum_{i=1}^n f(\gamma(t_i^*)) (\gamma(t_i) - \gamma(t_{i-1})).$$

Chopping up explanation:

- Partition $[a, b]$ into tiny intervals.
- Over each small interval, approximate the curve by a line segment: $\gamma(t_i) - \gamma(t_{i-1})$.
- Multiply by $f(\gamma(t_i^*))$ at a sample point.
- Sum over all intervals; refine partition to get the integral.

Why parameterization works: $\gamma'(t)\Delta t \approx \gamma(t_i) - \gamma(t_{i-1})$ for small Δt .

Example: Contour Integral

Unit circle: $\gamma(t) = e^{it}$, $t \in [0, 2\pi]$:

$$\int_{\gamma} z^n dz = \int_0^{2\pi} e^{int} i e^{it} dt = i \int_0^{2\pi} e^{i(n+1)t} dt$$

which vanishes when $n \neq -1$.

The Fundamental Computation

Let γ be the positively oriented circle

$$\gamma(t) = z_0 + re^{it}, \quad 0 \leq t \leq 2\pi.$$

Then

$$\gamma'(t) = ire^{it}, \quad \gamma(t) - z_0 = re^{it}.$$

Compute:

$$\int_{\gamma} \frac{1}{z - z_0} dz = \int_0^{2\pi} \frac{1}{re^{it}} (ire^{it}) dt.$$

Everything cancels:

$$= \int_0^{2\pi} i dt = i(2\pi) = \boxed{2\pi i}.$$

The entire effect is the winding of e^{it} once around the origin.

Green's Theorem

$$\oint_C (P \, dx + Q \, dy) = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

for positively oriented simple closed curve C and D its interior.

Interpretation: circulation around C = double integral of curl over D .

Proof of Green's Theorem (Idea)

Goal:

$$\oint_C (P dx + Q dy) = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

Step 1: Prove for a rectangle

Let $D = [a, b] \times [c, d]$ and assume P, Q have continuous partial derivatives.

Break the line integral into four sides.

Horizontal sides:

$$\int_a^b P(x, c) dx - \int_a^b P(x, d) dx = - \int_a^b \int_c^d \frac{\partial P}{\partial y} dy dx$$

(by the Fundamental Theorem of Calculus)

Vertical sides:

$$\int_c^d Q(b, y) dy - \int_c^d Q(a, y) dy = \int_c^d \int_a^b \frac{\partial Q}{\partial x} dx dy.$$

Finishing the Rectangle Case

Adding horizontal and vertical contributions:

$$\oint_C (P dx + Q dy) = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA.$$

Step 2: Extend to General Regions

- Decompose D into finitely many small rectangles.
- Apply the rectangle result to each.
- Interior boundary contributions cancel (opposite orientations).

Taking the limit as the mesh $\rightarrow 0$ yields the theorem for any region with piecewise smooth boundary.

Green's Theorem is FTC applied twice + cancellation.

Example (Green's Theorem)

$\oint_C (x^2 dx + y^2 dy)$ around unit circle:

$$\iint_D \left(\frac{\partial}{\partial x} y^2 - \frac{\partial}{\partial y} x^2 \right) dA = 0$$

Cauchy–Goursat Theorem (Section 6.3)

If f is analytic inside and on a simple closed contour γ ,

$$\int_{\gamma} f(z) dz = 0.$$

Allows deformation of contours for analytic functions.

From Green's Theorem to Cauchy–Goursat

Let $f(z) = u(x, y) + iv(x, y)$ be analytic with continuous partials.

Write the contour integral in real form:

$$\int_{\gamma} f(z) dz = \int_{\gamma} (u + iv)(dx + i dy).$$

Expand:

$$= \int_{\gamma} (u dx - v dy) + i \int_{\gamma} (v dx + u dy).$$

Apply Green's Theorem to each real integral.

$$\int_{\gamma} (u dx - v dy) = \iint_D (-v_x - u_y) dA$$

$$\int_{\gamma} (v dx + u dy) = \iint_D (u_x - v_y) dA.$$

If f is analytic, the Cauchy–Riemann equations hold:

The Fundamental Computation

Let γ be the positively oriented circle

$$\gamma(t) = z_0 + re^{it}, \quad 0 \leq t \leq 2\pi.$$

Then

$$\gamma'(t) = ire^{it}, \quad \gamma(t) - z_0 = re^{it}.$$

Compute:

$$\int_{\gamma} \frac{1}{z - z_0} dz = \int_0^{2\pi} \frac{1}{re^{it}} (ire^{it}) dt.$$

Everything cancels:

$$= \int_0^{2\pi} i dt = i(2\pi) = \boxed{2\pi i}.$$

The entire effect is the winding of e^{it} once around the origin.

The Winding Number

Let γ be a positively oriented, simple closed contour that avoids z_0 then

$$\int_{\gamma} \frac{1}{z - z_0} dz = 2\pi i \operatorname{Ind}(\gamma, z_0),$$

where $\operatorname{Ind}(\gamma, z_0)$ is the **winding number** of γ about z_0 .

Idea of Proof:

- Deform γ continuously to a circle around z_0 .
- The function $\frac{1}{z - z_0}$ is analytic away from z_0 .
- By Cauchy–Goursat, the integral is unchanged under deformation.
- We already computed the circle case: $2\pi i$.

If γ winds n times around z_0 , the integral is $2\pi in$.

The number of times the curve wraps around the z_0 .

Where Does Cauchy's Formula Come From?

Let f be analytic inside and on a simple closed contour γ , and let z_0 lie inside γ . Consider the function

$$g(z) = \frac{f(z) - f(z_0)}{z - z_0}.$$

Key observations:

- No singularity at z_0 because f is differentiable at z_0 ,

$$\lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0} = f'(z_0),$$

- $g(z)$ is analytic for $z \neq z_0$. So g is analytic inside γ .

By Cauchy–Goursat and the basic result.

$$\int_{\gamma} g(z) dz = \int_{\gamma} \frac{f(z)}{z - z_0} dz - \int_{\gamma} \frac{f(z_0)}{z - z_0} dz = \int_{\gamma} \frac{f(z)}{z - z_0} dz - 2\pi i f(z_0) = 0$$

- FTC links derivatives and integrals.
- Green's, Divergence, Stokes' theorems: local derivatives $\rightarrow$ global integrals.
- Riemann sum and sup/inf definitions give rigor.
- Complex contour integrals generalize line integrals; chopping explains the limit.
- Cauchy–Goursat: integral around closed loop vanishes for analytic functions.

Cauchy Integral Formula

Theorem. Let f be analytic on and inside a positively oriented simple closed contour γ . If z_0 lies inside γ , then

$$f(z_0) = \frac{1}{2\pi i} \int_{\gamma} \frac{f(z)}{z - z_0} dz.$$

Consequences:

- Values inside are determined by boundary values.
- Analytic functions are rigid.

Cauchy Formula for Derivatives

If f is analytic on and inside γ , then for $n \geq 1$:

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \int_{\gamma} \frac{f(z)}{(z - z_0)^{n+1}} dz.$$

Immediate Consequences:

- Analytic $\Rightarrow$ infinitely differentiable.
- Leads to Taylor series expansion:

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n.$$

Leibniz Rule (Differentiation Under the Integral)

Let

$$F(z) = \int_{\gamma} f(z, \zeta) d\zeta,$$

where $f(z, \zeta)$ is analytic in z and jointly continuous. Then

$$\frac{d}{dz} F(z) = \int_{\gamma} \frac{\partial}{\partial z} f(z, \zeta) d\zeta.$$

Higher derivatives can be computed by repeating the process.

This justifies differentiating Cauchy's formula.

Section 6.4: Consequences of Cauchy Formula

Theorem (Analytic $\Rightarrow C^\infty$).

If f is analytic in a region, then all derivatives exist and are analytic.

Theorem (Cauchy Estimates).

If $|f(z)| \leq M$ on $|z - z_0| = r$, then

$$|f^{(n)}(z_0)| \leq \frac{n!M}{r^n}.$$

Liouville's Theorem

Theorem.

If f is entire and bounded on $\mathbb{C}$, then f is constant.

Proof Sketch: Use Cauchy estimates and let $r \rightarrow \infty$:

$$|f'(z_0)| \leq \frac{M}{r} \rightarrow 0.$$

Hence $f'(z_0) = 0$ everywhere.

Fundamental Theorem of Algebra

Theorem. Every non-constant polynomial with complex coefficients has at least one complex root.

Proof Sketch: If a polynomial had no roots, $1/p(z)$ would be entire and bounded, so constant by Liouville — contradiction.

Maximum Modulus Principle

Theorem.

If f is analytic in a region D and attains a maximum of $|f|$ at an interior point, then f is constant.

Idea: Use Cauchy's formula and averaging over circles:

$$f(z_0) = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + re^{it}) dt.$$

Interior maxima force constancy.

Minimum Modulus Principle

If f is analytic and nonzero in D , then $|f|$ cannot attain a minimum at an interior point unless f is constant.

Follows from applying the Maximum Modulus Principle to $1/f$.

Morera's Theorem

Theorem (Morera).

Let f be continuous on a region D . Suppose that for every simple closed contour γ contained in D ,

$$\int_{\gamma} f(z) dz = 0.$$

Then f is analytic in D .

Interpretation:

- Cauchy–Goursat: analytic $\Rightarrow$ zero integrals.
- Morera: zero integrals $\Rightarrow$ analytic.

Vanishing contour integrals characterize analyticity.

Idea of Proof of Morera's Theorem

Fix $z_0 \in D$ and pick any contour $\gamma_{z_0 \rightarrow z}$ from z_0 to z that lies in D and define

$$F(z) = \int_{\gamma_{z_0 \rightarrow z}} f(\zeta) d\zeta,$$

$F(z)$ does not depend on path choice of path and

$$F'(z) = f(z).$$

and so f must be analytic.

Summary of Chapter 6

- Cauchy Integral Formula controls values and derivatives.
- Analytic $\Rightarrow$ infinitely differentiable.
- Cauchy estimates $\Rightarrow$ Liouville.
- Liouville $\Rightarrow$ Fundamental Theorem of Algebra.
- Maximum Modulus Principle controls geometry.
- Vanishing contour integrals characterize analyticity.

From Taylor to Laurent: Why We Need More

Recall: Taylor Series

If f is analytic in a disk $D(a, R)$, then

$$f(z) = \sum_{n=0}^{\infty} a_n (z - a)^n, \quad a_n = \frac{f^{(n)}(a)}{n!}.$$

But what if f is analytic in an annulus?

Suppose

f is analytic on $A = \{z : r < |z - a| < R\}$.

Examples:

$$\frac{1}{z}, \quad \frac{1}{z-1}, \quad \frac{1}{z^2+1}.$$

These functions are not analytic at $z = a$, but are analytic *around* $z = a$.

Taylor series cannot capture this behavior.

We need a series that allows *negative powers*.

Extending Taylor: The Laurent Idea

If f is analytic in an annulus

$$A = \{z : r < |z - a| < R\},$$

then we expand in *both directions*:

$$f(z) = \sum_{n=-\infty}^{\infty} c_n(z - a)^n.$$

Geometric Interpretation

Taylor: disk

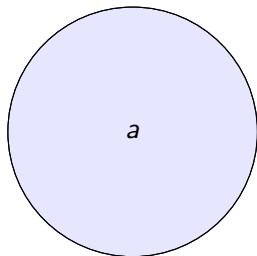
$$|z - a| < R$$

Laurent: annulus

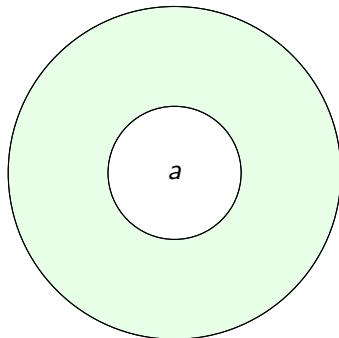
$$r < |z - a| < R$$

Negative powers describe the behavior near isolated singularities.

Geometry: Disk vs Annulus



Taylor: $|z - a| < R$



Laurent: $r < |z - a| < R$

Taylor: analytic *at* the center.

Laurent: analytic *around* the center.

Worked Example: Two Regions

Let

$$f(z) = \frac{1}{z(z-1)}.$$

Partial fractions:

$$\frac{1}{z(z-1)} = -\frac{1}{z} + \frac{1}{z-1}.$$

Region 1: $|z| < 1$

$$\frac{1}{z-1} = -\frac{1}{1-z} = -\sum_{n=0}^{\infty} z^n.$$

Thus

$$f(z) = -\frac{1}{z} - \sum_{n=0}^{\infty} z^n.$$

Laurent series with principal part $-\frac{1}{z}$.

Worked Example: Exterior Region

Region 2: $|z| > 1$

$$\frac{1}{z-1} = \frac{1}{z} \frac{1}{1 - \frac{1}{z}} = \frac{1}{z} \sum_{n=0}^{\infty} \frac{1}{z^n} = \sum_{n=1}^{\infty} \frac{1}{z^n}.$$

Therefore

$$f(z) = -\frac{1}{z} + \sum_{n=1}^{\infty} \frac{1}{z^n} = \sum_{n=2}^{\infty} \frac{1}{z^n}.$$

Now the singularity at 0 disappears.

Region determines the expansion.

Preview: Residues

If

$$f(z) = \sum_{n=-\infty}^{\infty} c_n(z-a)^n,$$

then the coefficient

$$c_{-1}$$

is called the **residue** of f at a .

Residues determine contour integrals:

$$\oint_{\gamma} f(z) dz = 2\pi i \cdot (\text{sum of residues inside } \gamma).$$

Laurent series unlock powerful integration tools.

Classification of Isolated Singularities

Suppose f has Laurent expansion near a :

$$f(z) = \sum_{n=-\infty}^{\infty} c_n(z-a)^n.$$

Removable:

$$c_{-n} = 0 \text{ for all } n \geq 1.$$

Pole of order m :

$$c_{-m} \neq 0, \quad c_{-n} = 0 \text{ for } n > m.$$

Essential singularity:

infinitely many negative powers.

The principal part tells the whole story.